MILITARY SPECIFICATION SHEET

SCANNER, MECHANICAL, MX-10717(V)4/U

This specification is approved for use by USACECOM, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification sheet covers the Mechanical Scanner, MX-10717(V)4/U and augments MIL-S-49301. The MX-10717(V)4/U is used on the Night Chaparral FLIR Subsystem.

1.2 Purpose. The purpose of this specification sheet is to modify or expand the contents of MIL-S-49301 to describe the Scanner Assembly. All the MIL-S-49301 contents apply except as specifically modified herein. Paragraph numbers directly correlate between MIL-S-49301 and this specification sheet.

2. APPLICABLE DOCUMENTS

2.1 Order of precedence. In the event of a conflict between the text of this specification sheet and the references cited herein, the text of this specification sheet shall take precedence.

SPECIFICATIONS

MILITARY

MIL-S-49301	- Scanner, Mechanical, MX-10717(V)/U, General Specification for

DRAWINGS

USA COMMUNICATIONS-ELECTRONICS COMMAND

54490-5006243-4	- Scanner, Mechanical, MX-10717(V)4/U
30063-SM-D-773946	- Scan-Interlace, PL-1408/UA

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: HQ, USA Communications-Electronics Command, ATTN: AMSEL-ED-TO, Fort Monmouth, NJ 07703, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5855

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3. REQUIREMENTS

3.1 Description. The MX-10717(V)4/U Scanner, herein known as the Scanner, shall consist of the MIL-S-49301, MX-10717(V)/U Scanner adjusted for 60 Hz operation.

3.2 Construction. The MX-10717(V)4/U shall be constructed in accordance with drawing 54490-5006243-4.

3.2.1 Weight. Weight of the Scanner shall be 1.55 pounds maximum.

3.6 Performance characteristics. The characteristics specified herein shall apply as defined in 3.1 and 3.6.1 except 4.6.8, 4.7.4, and 4.7.5 applies with the phase shift lens, counterweight and associated hardware removed. Operation shall be at 60 Hz using the Scan-Interlace, PL-1408/UA (SM-D-773946).

3.6.1 Scan angle and frequency.

3.6.1.1 Variable scan frequency. Not applicable.

3.6.1.2 Specific scan frequency (60 Hz). Scan frequency of the Scanner shall be adjusted to 60.0, ±1.0 Hz. Scan frequency drift shall not be greater than ±6.0 Hz from the frequency at ambient setting (see MIL-S-49301, 4.4) when exposed to high and low temperature environmental conditions (see MIL-S-XXXXX).

3.6.1.3 Synchronous scan frequency. The scanner shall synchronize with a stable external synchronization source, such as the TV camera vertical sync pulse. Scan frequency shall be stable within 1 Hz of the input sync pulse frequency.

3.6.1.4 Scan angle. The scan angle shall be 12.2, ±0.5 degrees.

3.6.2 Interlace. The scanner shall operate in a 2:1 interlace mode. The interlace switching time shown in figure 1 of MIL-S-49301 shall not be greater than 2.5 milliseconds (ms).

3.6.3 Scan direction. The Scanner shall show the waveform shapes and phase relationships in figure 1 while operating in the 2:1 rear interlace mode.

3.6.4 Power consumption. Power required to operate the Scanner shall not exceed 7.5 watts.

3.6.5 Scan efficiency. The Scanner scan efficiency shall be a minimum of 70 percent.

3.6.6 Scan jitter in azimuth. Jitter time shall be 12.5 microseconds maximum.

4. QUALITY ASSURANCE PROVISIONS

4.6 Test methods. Tests shall be performed utilizing the SM-D-773946 Scan-Interlace with the Scanner with a scan angle of 12.2, ±0.5 degrees at the 60.0, ±1.0 Hz scan rate.

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4.6.1 Scan frequency.

4.6.1.1 Variable scan frequency. Not applicable.

4.6.1.2 Specific scan frequency (60 Hz). The Scanner shall be connected to the Scan-Interlace, SM-D-773946, and mounted to a suitable fixture. Power shall be applied and the output of the azimuth (scan) transducer (Scan-Interlace SM-D-773946) shall be monitored with a calibrated frequency counter. The output of the Scan-Interlace (SM-D-773946) shall be adjusted to obtain the specific scan frequency. Failure to meet the requirements of 3.6.1.2 shall constitute failure of this test.

4.6.1.3 Synchronous scan frequency. The Scanner shall be connected to Scan-Interlace SM-D-773946 and mounted in a suitable fixture. Synchronization from an external stable source (such as a TV camera vertical sync pulse) and power shall be applied and the output of the azimuth (scan) transducer (Scan-Interlace SM-D-773946) shall be monitored with a calibrated frequency counter. Failure to meet the requirements of 3.6.1.3 shall constitute failure of this test.

4.6.1.4 Scan angle. See MIL-S-49301.

4.6.2 Interlace. See MIL-S-49301.

4.6.3 Scan direction. With the Scanner operating, the outputs of the azimuth and interlace position transducers (Scan-Interlace SM-D-773946) shall be observed with an oscilloscope. The output waveforms shall show the shape and phase relationships shown in figure 1. Failure to meet the requirements of 3.6.3 shall constitute failure of this test.

4.6.4 Power consumption. See MIL-S-49301.

4.6.5 Scan efficiency. See MIL-S-49301.

4.6.6 Scan jitter in azimuth. See MIL-S-49301.

6. NOTES

6.1 Intended use. The Scanner is intended for use with the Night Chaparral FLIR Subsystem.

6.2 Interchangeability. The MX-10717(V)4/U scanner built and tested in accordance with this specification and applicable drawings may be used in place of the MX-9872(V)4/UA scanner. The MX-9872(V)4/UA may not be used in place of the MX-10717(V)4/U scanner.

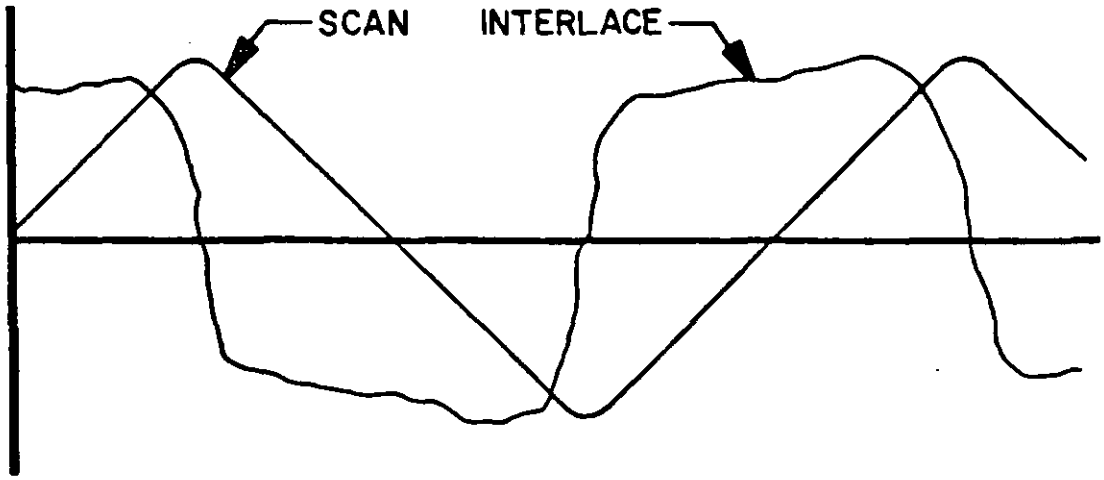
6.3 Subject term (keyword) listing.
FLIR
Sensor
Scanner

CUSTODIAN
Army - CR

PREPARING ACTIVITY
Army - CR
Project 5855-A290-4

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2:1 REAR POSITION
INTERLACE LEADS SCAN APPROX. 90°



NOTES:

PL-1408/UA- CONNECT P1-33 TO P1-34 AND P1-4 TO P1-17

OSCILLOSCOPE- CONNECT CHANNEL 1 (SCAN POSITION) HIGH INPUT TO J3 (OR J2-7) AND LOW SCOPE TO GND

- CONNECT CHANNEL 2 (SCAN INTERLACE) HIGH INPUT TO J6 (OR J1-9) AND LOW SCOPE TO GND

FIGURE 1. INTERLACE TRANSDUCER AND AZIMUTH POSITION TRANSDUCER OUTPUTS.

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1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-S-49301/4(CR)	Scanner, Mechanical, MX-10717(V)4/U

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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